

A Practical Guide to Learning rate neural network

TOPIC -- LEARNING RATE NEURAL NETWORK

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

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Large-scale automatic speech recognition is the first and most convincing successful case of deep learning. LSTM RNNs can learn "Very Deep Learning" tasks that involve multi-second intervals containing speech events separated by thousands of discrete time steps, where one time step corresponds to about 10 ms. LSTM with forget gates is competitive with traditional speech recognizers on certain tasks. The initial success in speech recognition was based on small-scale recognition tasks based on TIMIT. The data set contains 630 speakers from eight major dialects of American English, where each speaker reads 10 sentences. Its small size lets many configurations be tried. More importantly, the TIMIT task concerns phone-sequence recognition, which, unlike word-sequence recognition, allows weak phone bigram language models. This lets the strength of the acoustic modeling aspects of speech recognition be more easily analyzed. The error rates listed below, including these early results and measured as percent phone error rates (PER), have been summarized since 1991.

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Before 1980 There are two types of artificial neural network (ANN): feedforward neural network (FNN) or multilayer perceptron (MLP) and recurrent neural networks (RNN). RNNs have cycles in their connectivity structure, whereas FNNs do not. In the 1920s, Wilhelm Lenz and Ernst Ising created the Ising model which is essentially a non-learning RNN architecture consisting of neuron-like threshold elements. In 1972, Shun'ichi Amari made this architecture adaptive. His learning RNN was republished by John Hopfield in 1982. Other early recurrent neural networks were published by Kaoru Nakano in 1971. Already in 1948, Alan Turing produced work on "Intelligent Machinery" that was not published in his lifetime, containing "ideas related to artificial evolution and learning RNNs". Frank Rosenblatt (1958) proposed the perceptron, an MLP with 3 layers: an input layer, a hidden layer with randomized weights that did not learn, and an output layer. He later published a 1962 book that also introduced

variants and computer experiments, including a version with four-layer perceptrons "with adaptive preterminal networks" where the last two layers have learned weights (here he credits H. D. Block and B. W. Knight). The book cites an earlier network by R. D. Joseph (1960) "functionally equivalent to a variation of" this four-layer system (the book mentions Joseph over 30 times). Should Joseph therefore be considered the originator of proper adaptive multilayer perceptrons with learning hidden units? Unfortunately, the learning algorithm was not a functional one, and fell into oblivion. The first working deep learning algorithm was the Group method of data handling, a method to train arbitrarily deep neural networks, published by Alexey Ivakhnenko and Lapa in 1965. They regarded it as a form of polynomial regression, or a generalization of Rosenblatt's perceptron to handle more complex, nonlinear, and hierarchical relationships. A 1971 paper described a deep network with eight layers trained by this method, which is based on layer by layer training through regression analysis. Superfluous hidden units are pruned using a separate validation set. Since the activation functions of the nodes are Kolmogorov-Gabor polynomials, these were also the first deep networks with multiplicative units or "gates". The first deep learning multilayer perceptron trained by stochastic gradient descent was published in 1967 by Shun'ichi Amari. In computer experiments conducted by Amari's student Saito, a five layer MLP with two modifiable layers learned internal representations to classify non-linearly separable pattern classes. Subsequent developments in hardware and hyperparameter tunings have made end-to-end stochastic gradient descent the currently dominant training technique. In 1969, Kunihiko Fukushima introduced the ReLU (rectified linear unit) activation function. The rectifier has become the most popular activation function for deep learning. Deep learning architectures for convolutional neural networks (CNNs) with convolutional layers and downsampling layers began with the Neocognitron introduced by Kunihiko Fukushima in 1979, though not trained by backpropagation. Backpropagation is an efficient application of the chain rule derived by Gottfried Wilhelm Leibniz in 1673 to networks of differentiable nodes. The terminology "back-propagating errors" was actually introduced in 1962 by Rosenblatt, but he did not know how to implement this, although Henry J. Kelley had a continuous precursor of backpropagation in 1960 in the context of control theory. The modern form of backpropagation was first published in Seppo Linnainmaa's master thesis (1970). G.M. Ostrovski et al. republished it in 1971. Paul Werbos applied backpropagation to neural networks in 1982 (his 1974 PhD thesis, reprinted in a 1994 book, did not yet describe the algorithm). In 1986, David E. Rumelhart et al. popularised backpropagation but did not cite the original work.

The deep learning revolution started around CNN- and GPU-based computer vision. Although CNNs trained by backpropagation had been around for decades and GPU implementations of NNs for years, including CNNs, faster implementations of CNNs on GPUs were needed to progress on computer vision. Later, as deep learning becomes widespread, specialized hardware and algorithm optimizations were developed specifically for deep learning. A key advance for the deep learning revolution was hardware advances, especially GPU. Some early work dated back to 2004. In 2009, Raina, Madhavan, and Andrew Ng reported a 100M deep belief network trained on 30 Nvidia GeForce GTX 280 GPUs, an early demonstration of GPU-based deep learning. They reported up to 70 times faster training. In 2011, a CNN named DanNet by Dan Ciresan, Ueli Meier, Jonathan Masci, Luca Maria Gambardella, and Jürgen Schmidhuber achieved for the first time superhuman performance in a visual pattern recognition contest, outperforming traditional methods by a factor of 3. It then won more contests. They also showed how max-pooling CNNs on GPU improved performance significantly. In 2012, Andrew Ng and Jeff Dean created an FNN that learned to recognize higher-level concepts, such as cats, only from watching unlabeled images taken from YouTube videos. In October 2012, AlexNet by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton won the large-scale ImageNet competition by a significant margin over shallow machine learning methods. Further incremental improvements included the VGG-16 network by Karen Simonyan and Andrew Zisserman and Google's Inceptionv3. The success in image classification was then extended to the more challenging task of generating descriptions (captions) for images, often as a combination of CNNs and LSTMs. In 2014, the state of the art was training "very deep neural network" with 20 to 30 layers. Stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In 2015, two techniques were developed to train very deep networks: the highway network was published in May 2015, and the residual neural network (ResNet) in Dec 2015. ResNet behaves like an open-gated Highway Net. Around the same time, deep learning started impacting the field of art. Early examples included Google DeepDream (2015), and neural style transfer (2015), both of which were based on pretrained image classification neural networks, such as VGG-19. Generative adversarial network (GAN) by (Ian Goodfellow et al., 2014) (based on Jürgen Schmidhuber's principle of artificial curiosity) became state of the art in generative modeling during 2014-2018 period. Excellent image quality is achieved by Nvidia's StyleGAN (2018) based on the Progressive GAN by Tero Karras et al. Here the GAN generator is grown from small to large scale in a pyramidal fashion. Image generation by GAN reached popular success, and provoked discussions concerning deepfakes. Diffusion models (2015) eclipsed GANs in generative modeling since then, with systems such as DALL·E 2 (2022)

and Stable Diffusion (2022). In 2015, Google's speech recognition improved by 49% by an LSTM-based model, which they made available through Google Voice Search on smartphone. Deep learning is part of state-of-the-art systems in various disciplines, particularly computer vision and automatic speech recognition (ASR). Results on commonly used evaluation sets such as TIMIT (ASR) and MNIST (image classification), as well as a range of large-vocabulary speech recognition tasks have steadily improved. Convolutional neural networks were superseded for ASR by LSTM. but are more successful in computer vision. Yoshua Bengio, Geoffrey Hinton and Yann LeCun were awarded the 2018 Turing Award for "conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing".